

MYSTERY OF TRAIN TRACKS: A FULL-STACK INTERACTIVE WEB GAME

(REFINED VER.)

TEAM 06

Chan Hio Tou (Collins) - DC325713

Xu WuSiyuan (Bruce) - DC327035

Uriel Wuli - DC227352

Tan Pak Long (John) dc226991

Seki Tomoki, DC227171

Chen Joaquin Antonio, DC226973

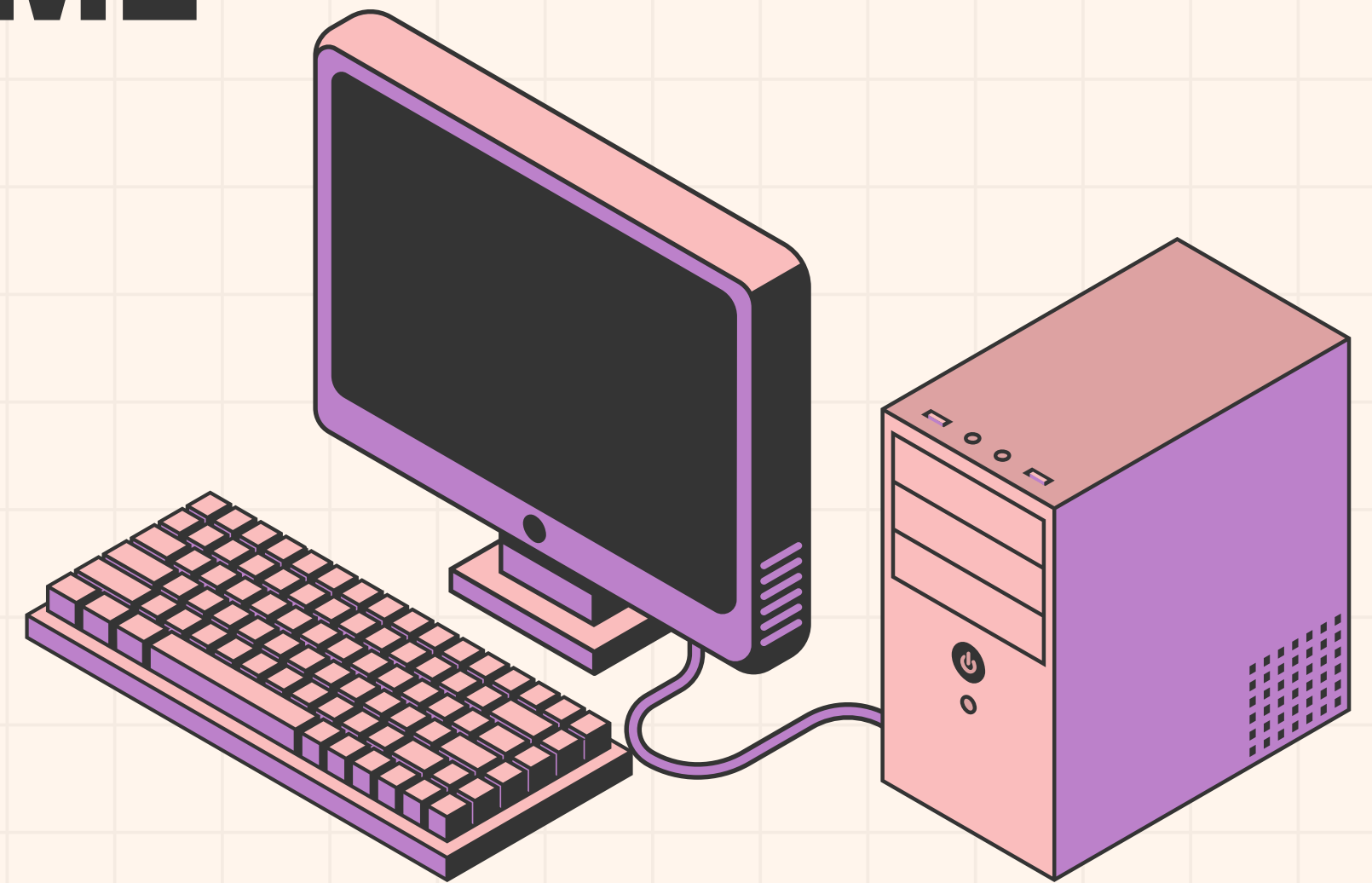
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Speaker: John, Tan Pak Long



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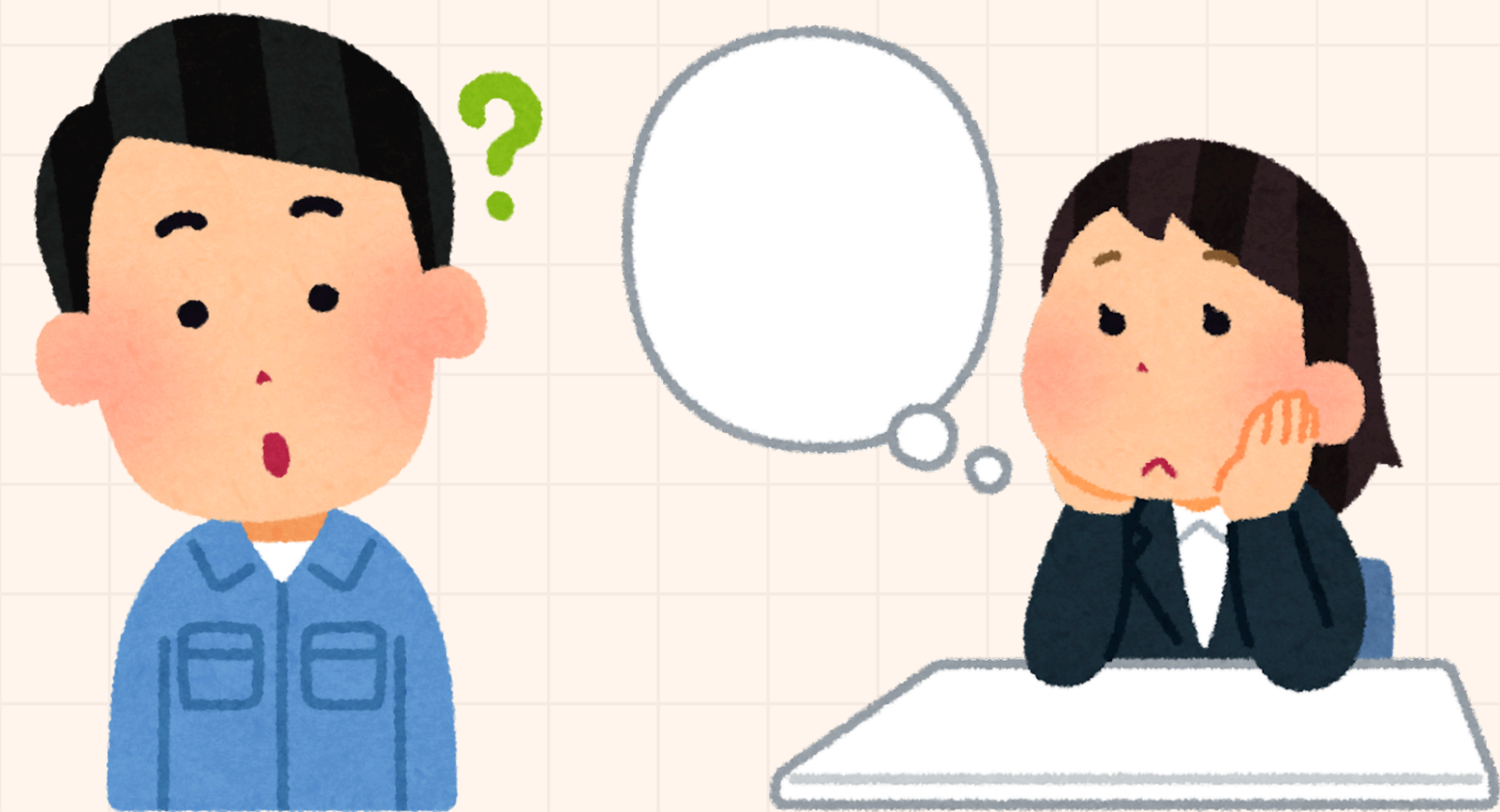
Speaker: John, Tan Pak Long

- Project Title
 - Context
 - Goals
 - Services & Data Requirements
 - Tasks and Tools
 - Our Work Distribution + Progress
- Deliverables
 - Criteria for Success
 - Evaluation
 - Project Learning
 - Demo Presentation



PROJECT TITLE

Mystery of Train Tracks: A Full-Stack Interactive Web Game



A guiding question...

How can a responsive, secure, and feature-rich web application deliver both engaging gameplay and robust user services in a puzzle-based environment?



CONTEXT

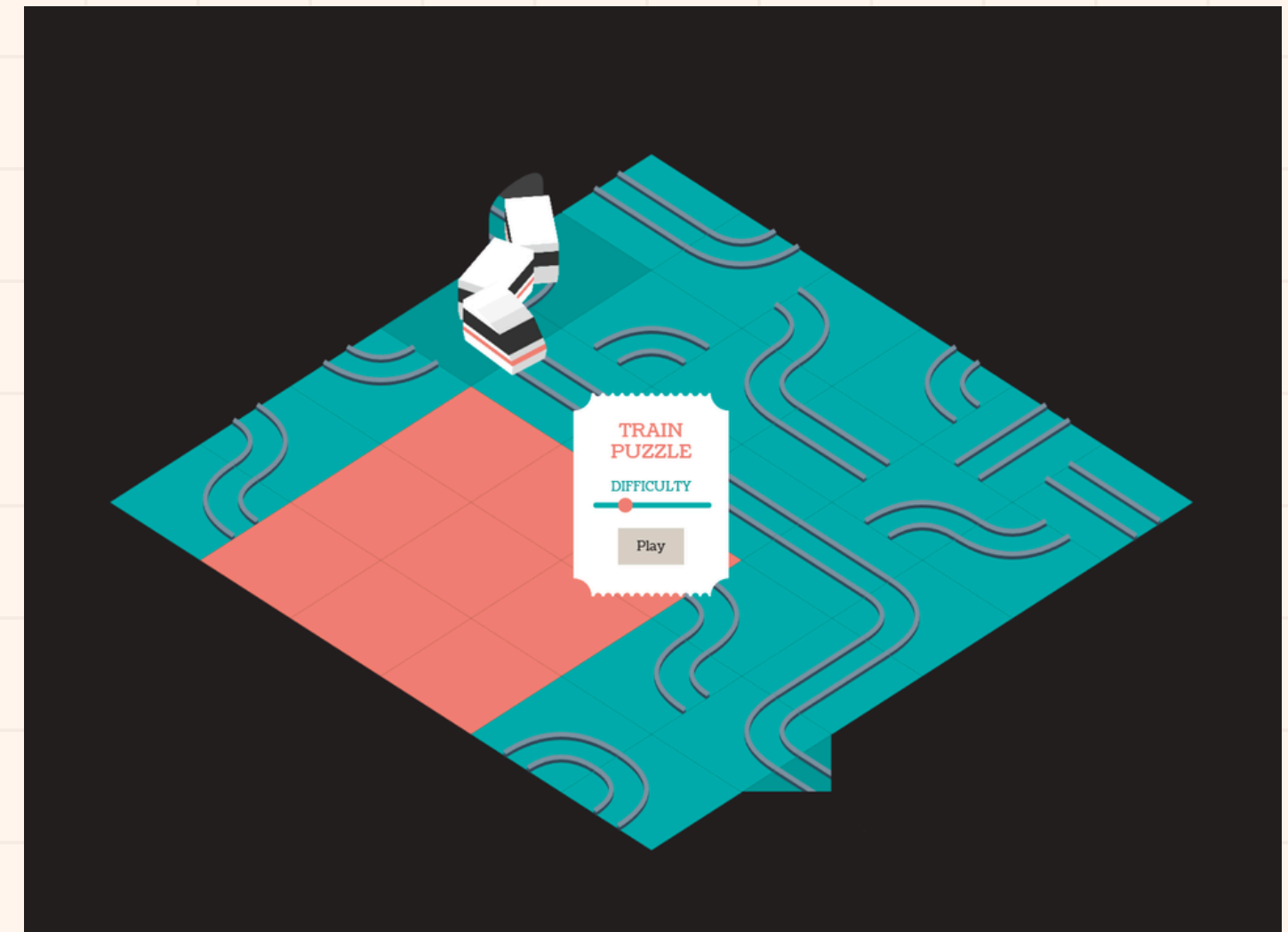
Mystery of Train Tracks: A Full-Stack Interactive Web Game

Our project, titled "The Mystery of the Railway: A Full-Stack Interactive Online Game," addresses the core challenge of creating a responsive, secure, and feature-rich web application that simultaneously delivers an engaging gaming experience and robust user service within a puzzle environment.

In this context, our project is not merely a course assignment but a comprehensive practice of full-stack development. It fully integrates front-end responsive design, back-end API and database management, and user authentication and security mechanisms, comprehensively covering the learning outcomes of the course. Team members will collaborate, each responsible for different aspects: UI design, authentication and search functionality, database schema and logging system, and will employ agile methodologies for iterative development and testing.

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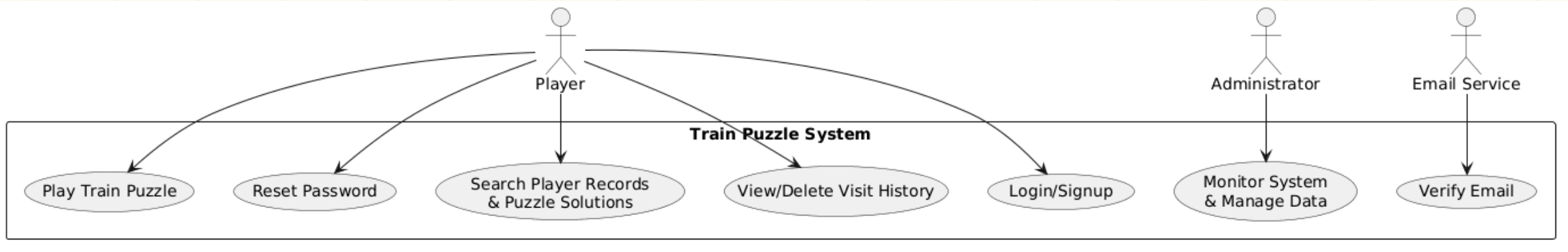
source: <https://codepen.io/johan-tirholm/pen/xoNPzN>

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Game represented in UML Use Case Diagram.

(shows the relationship between actors and system functions, clarifying responsibilities and interactions)



Actors

Player

(interacts with game, login/signup, search, history)

Administrator

(manages user data, oversees system)

Email Service

(handles verification and password reset)

Use Cases

Play Train Puzzle

Login/Signup

Verify Email

Reset Password

Search Player Records/Puzzle Solutions

View/Delete Visit History

Monitor system and manage data



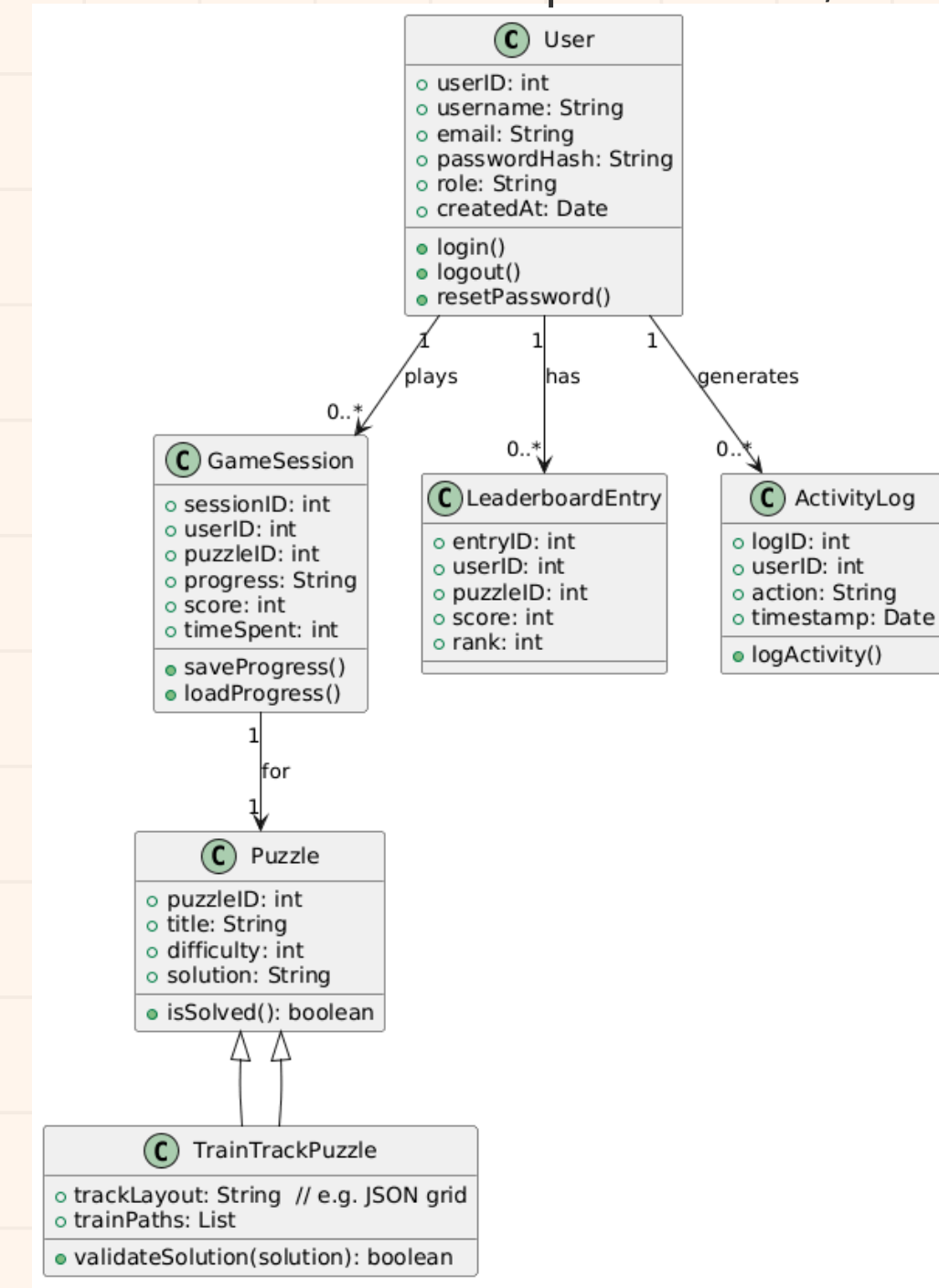
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Game represented in UML Class Diagram.

- User (User Class): Stores user information and provides login, logout, and password reset functions.
- GameSession (Game Session Class): Records the player's progress, score, and time spent on puzzles.
- Puzzle (Puzzle Class): Defines the puzzle's title, difficulty, and solution method.
- TrainTrackPuzzle (Train Track Puzzle Class): Inherits from Puzzle, adds track layout and train paths, and can verify solutions.
- LeaderboardEntry (Leaderboard Entry Class): Saves player scores and rankings.
- ActivityLog (Activity Log Class): Records user actions and timestamps.



CONTEXT

Game represented in BPMN workflow.

The workflow consists of five main components:

1. User Registration/Login

– User submits credentials → System validates → Email verification triggered → Account activated

2. Gameplay Flow

– User starts Train Puzzle → Game logic executes → Results stored in database → Records updated

3. Search Service

– User enters query → API fetches results from database → Results displayed in UI

4. Visit History Tracking

– Each page visit logged → Stored in history table → User can view/delete history

5. Admin Oversight

– Admin dashboard → Access to aggregated player data → Audit logs maintained

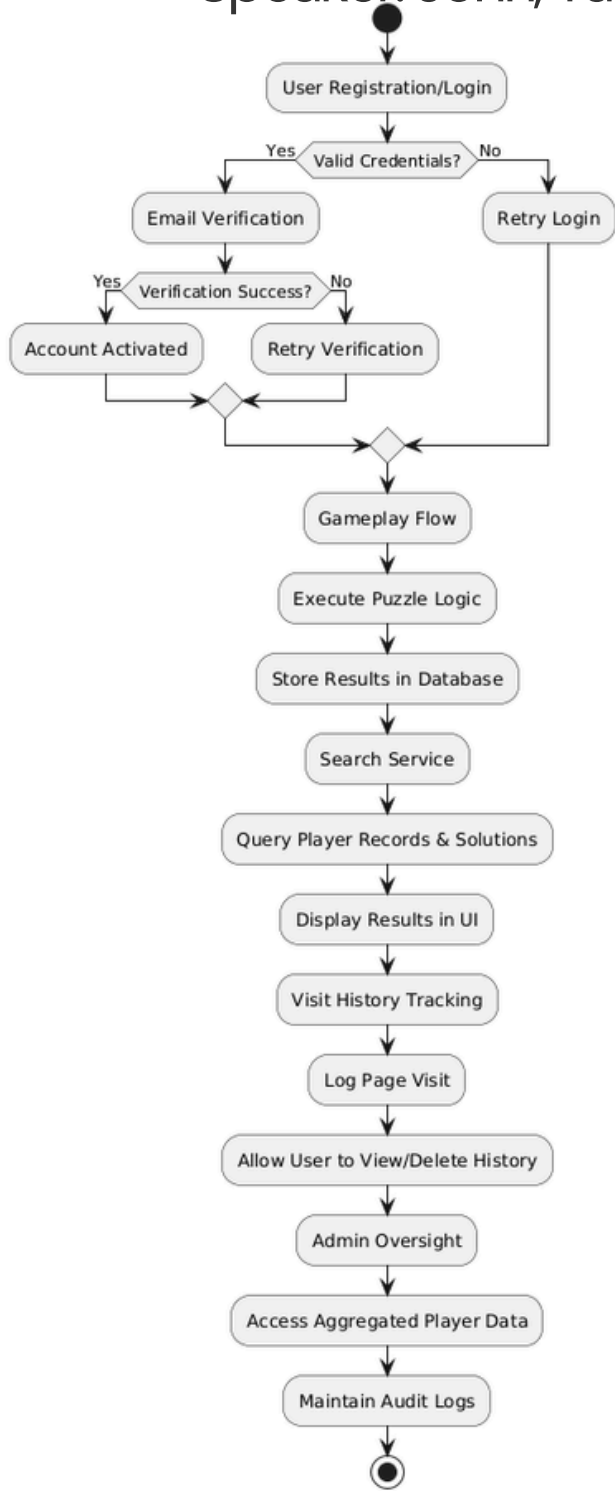
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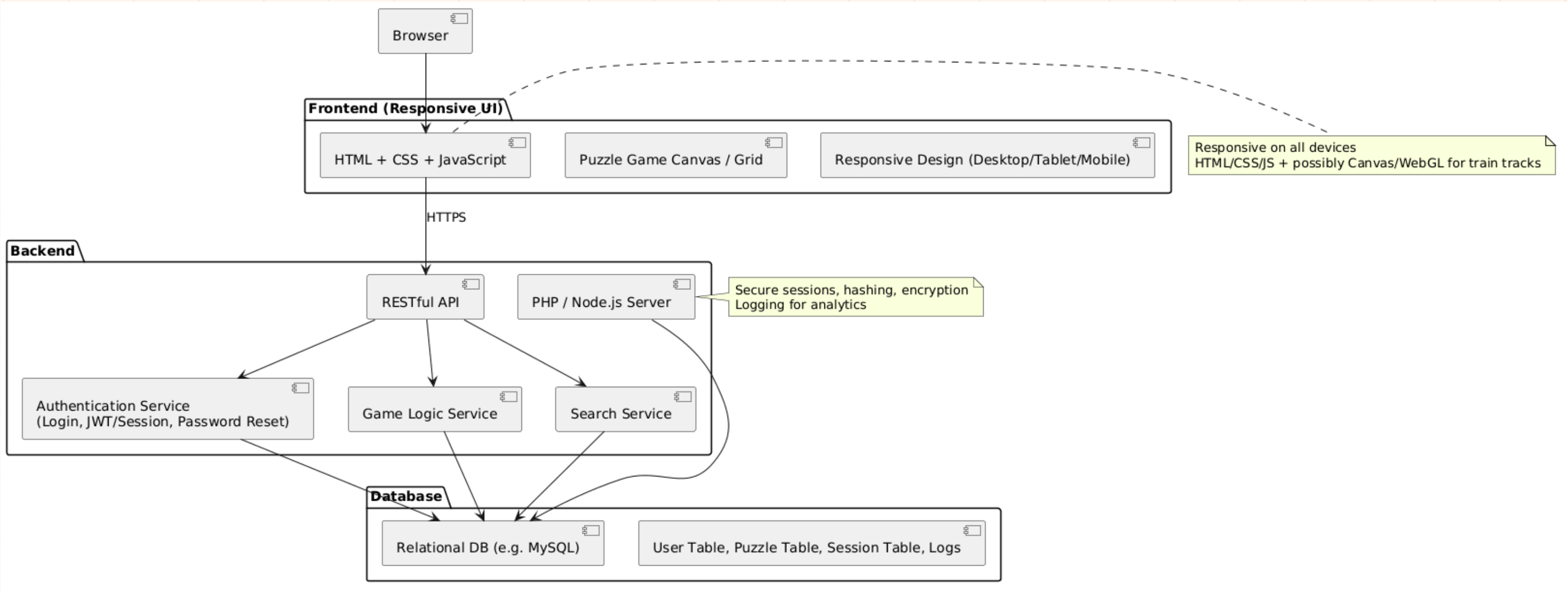
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CONTEXT

Speaker: John, Tan Pak Long

Overall description: Front-end / Back-end



GOALS - FINAL CONCEPT

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We refined our ultimate goals as follows:



- Build a complete, full-stack interactive game website for a **smooth gaming experience** on various devices
- Provide **secure and reliable user services** (e.g. registration, login, email verification, password reset, etc.)
- Support **data management** and analysis (e.g. player game records, puzzle results, and access history)
- Achieve **team collaboration and agile development**
- Demonstrate **professional capabilities**



GOALS -

FURTHER CODE IMPLEMENTATION GOALS (TECHNOLOGY DEPLOYMENT)

Frontend Goals

- Implement responsive design.
- Implement the game interface (tracks, trains, puzzle logic), menus, and player result display.
- Ensure the UI adapts to different devices.

Backend Goals

- Build an API service.
- Implement registration/login, email verification, password reset, and session management.
- Provide a search interface (player records, puzzle solutions).

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GOALS -

FURTHER CODE IMPLEMENTATION GOALS (TECHNOLOGY DEPLOYMENT)

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Database Goals

- Define user, game history, and access log schemas.
- Implement data storage, querying, and log tracing.
- Optimize query performance using indexes and caching.

Security Goals

- Protect user passwords using hash algorithms.
- Implement encrypted communication and secure sessions.
- Provide user privacy settings (such as deleting browsing history).

Deployment Goals

- Deploy the application.
- Ensure a smooth deployment by leveraging GitHub CI/CD automated workflows.
- Ensure the website is accessible online and supports multi-user concurrency.



SERVICES & DATA REQUIREMENTS

Speaker: Bruce, Xu WuSiyuan

SERVICES

Authentication

- User accounts, email verification, password reset

Search

- Query player records and puzzle solutions

Visiting History

- Log visited pages with timestamps

Game

- Puzzle logic, scoring, difficulty progression / customization

DATA REQUIREMENTS

User Table

- accounts, credentials, verification status

Game Records

- scores, attempts

History Table

- visited URLs, timestamps

Puzzle Solutions



TASKS AND TOOLS

Overview - Tasks

Speaker: Bruce, Xu WuSiyuan

SYSTEM DESIGN



Define system architectures, create UI wireframes and database E-R diagrams.

IMPLEMENTATION

Build responsive UI, develop authentication and search APIs, design database schemas.



TESTING & DEPLOYMENT

Conduct unit testing and usability testing, deploy the application to a PHP-enabled cloud platform.



TASKS AND TOOLS

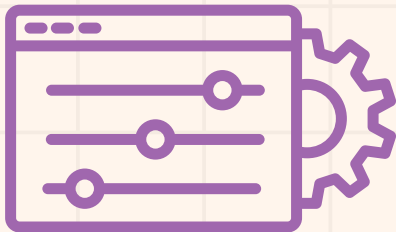
Overview - Tools

Speaker: Bruce, Xu WuSiyuan



FRONT-END

Of course, the three important components that make a functional website UI...



BACK-END

Frameworks required:



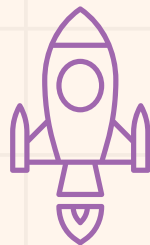
DATABASE

Primarily using MySQL for strict data structure, complex analytical queries, and absolute transactional reliability.



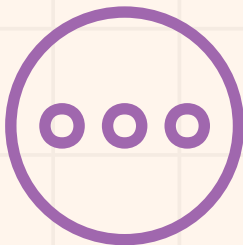
TASKS AND TOOLS

Overview - Tools



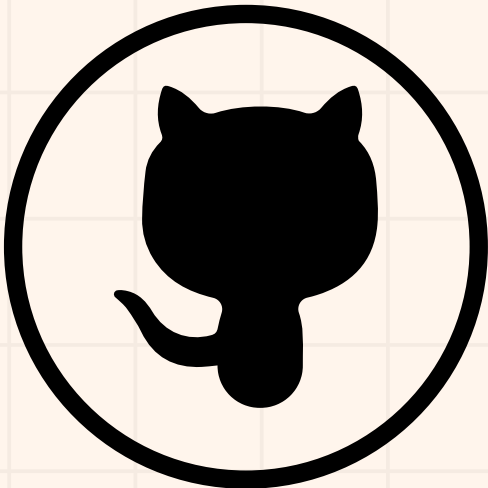
DEPLOYEMENT

We plan to deploy our work through Alibaba Cloud.



SUPPLEMENTS

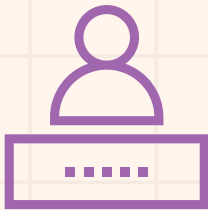
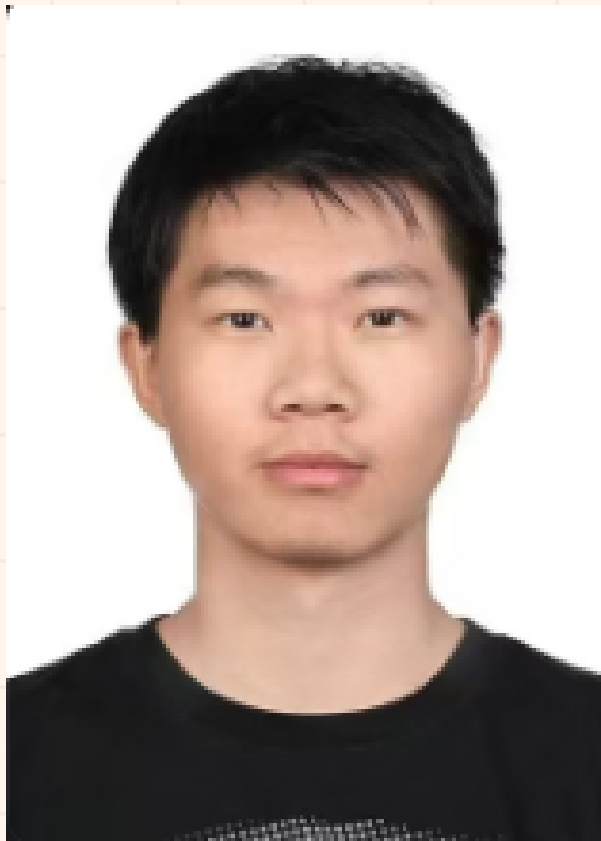
GitHub for seamless automation, especially of its consistent environment provided by GitHub actions.



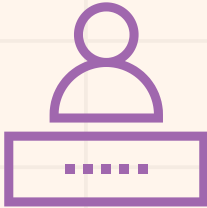
TASK DISTRIBUTION

PAIR 20 - FRONT-END INTERACTIVE DESIGN

Speaker: Bruce, Xu WuSiyuan



XU WUSIYUAN
(BRUCE)



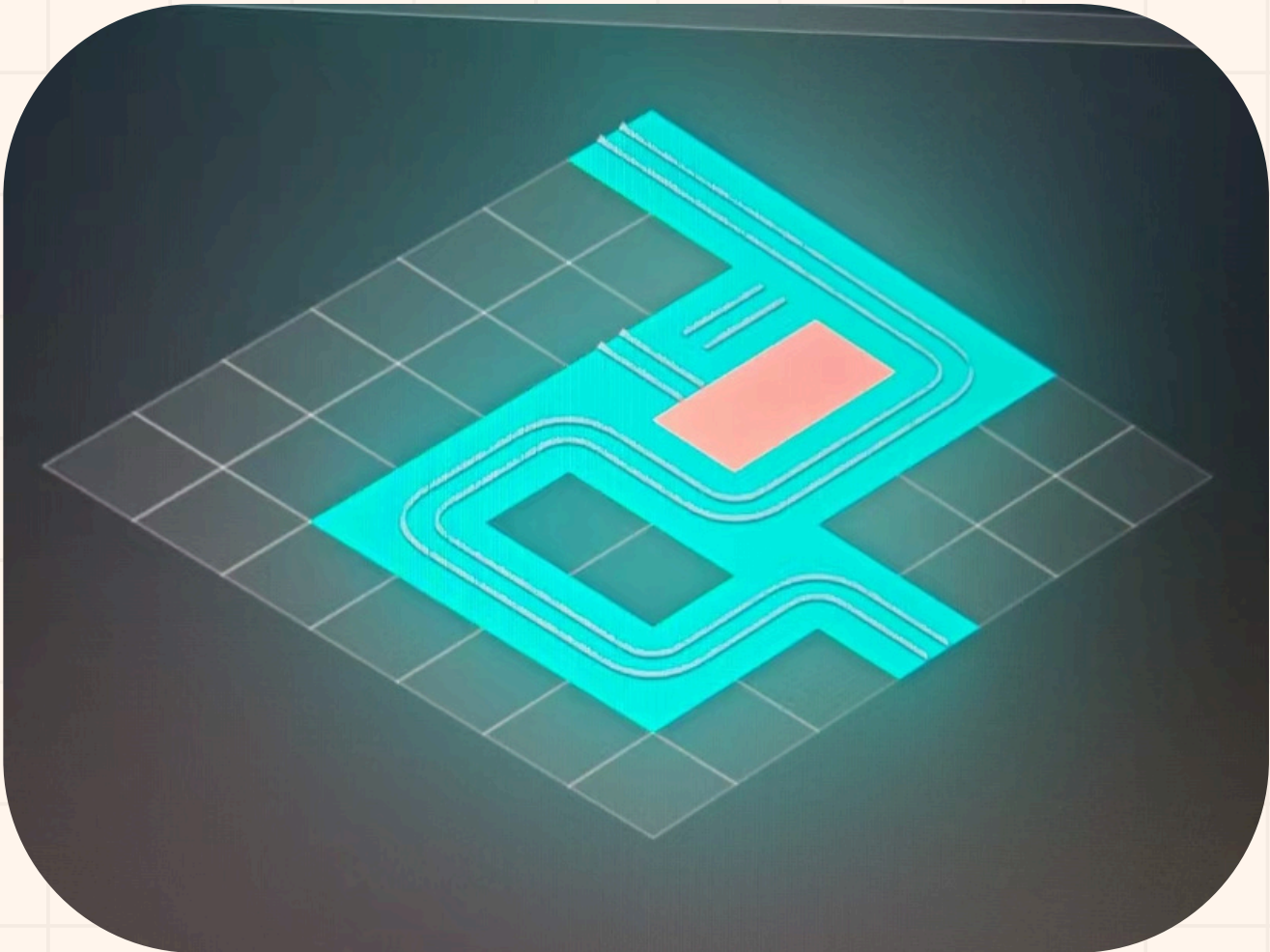
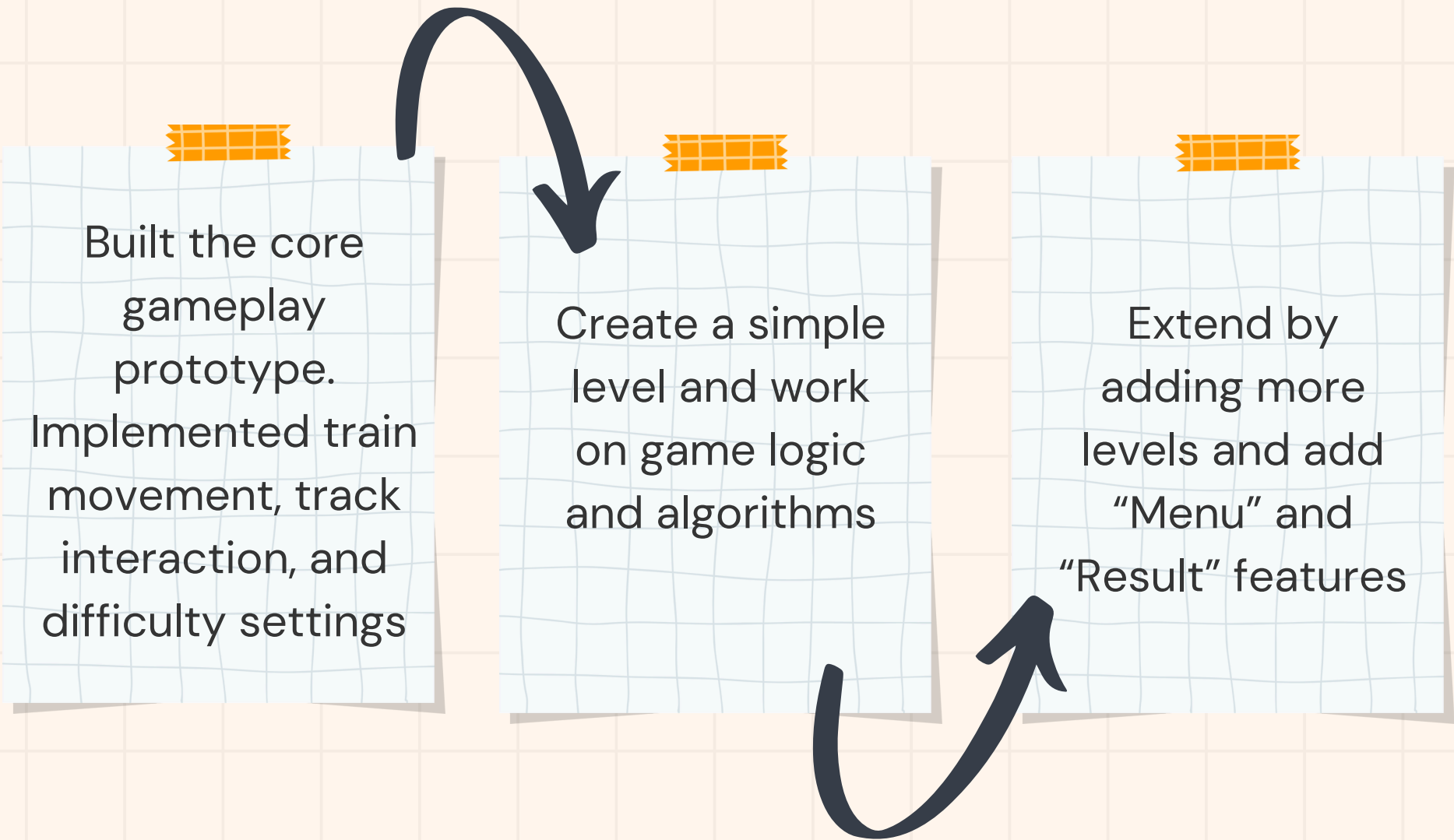
CHAN HIO TOU
(COLLINS)

Responsible for building responsive UI for puzzle game and user dashboard.

TASK DESCRIPTION

Front-end Interactive Design

Bruce and Collins are in charge of building a responsive User Interface for the puzzle game, including gameplay, menu and results.

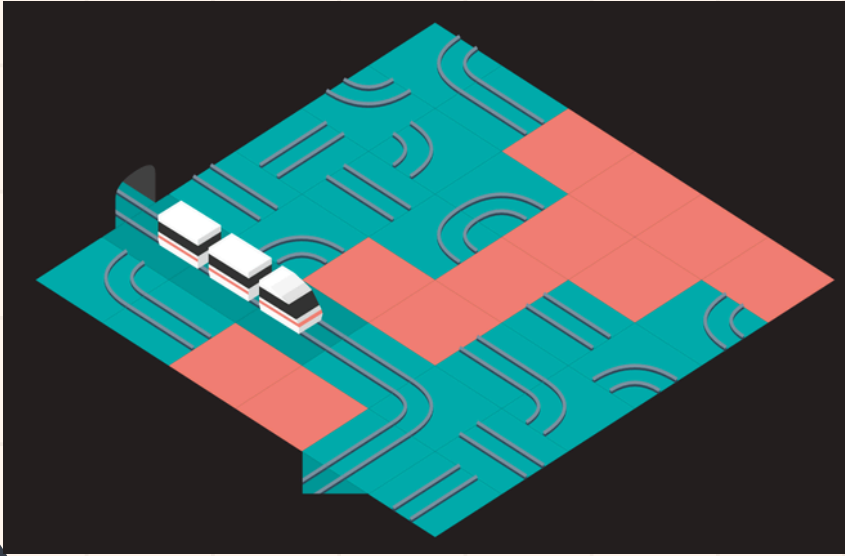


A simulated puzzle board by Bruce.



WORK PROGRESS

Front-end Interactive Design



Built the core gameplay prototype. Implemented train movement, track interaction, and difficulty settings



Create different level and work on game logic and algorithms



Connecting the game with back-end data storage add "Menu" and "Result" features



TASK DISTRIBUTION

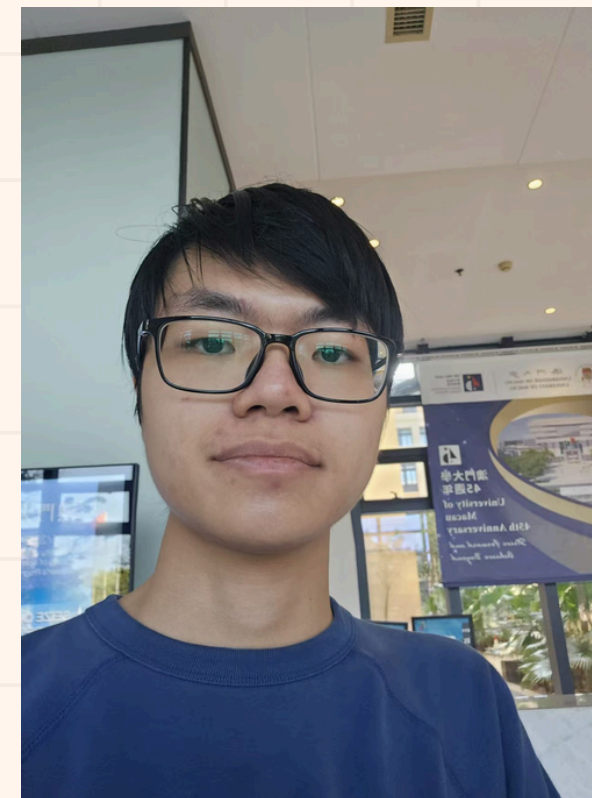
PAIR 11 - BACK-END DEVELOPMENT

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Speaker: Uriel Wuli



URIEL WULI



TAN PAK LONG,
JOHN

Responsible for implementing APIs for authentication, search and history tracking.

TASK DESCRIPTION

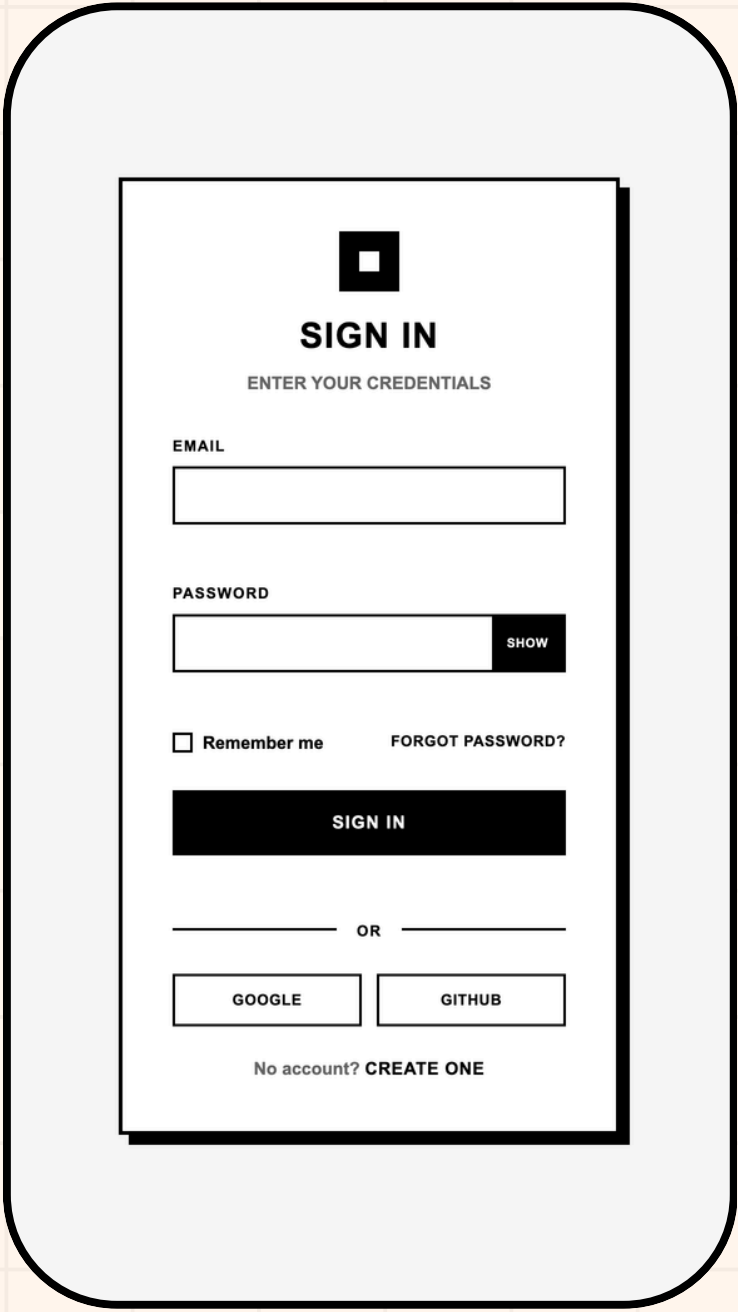
Back-end Development

Workflow Plan:

- Design an engaging, game-themed authentication interface utilizing HTML, CSS, and JavaScript to ensure a seamless and responsive user experience.
- Utilize PHP to develop server-side scripts that handle critical operations, including session management, secure password modification, and automated email dispatching.
- Establish a secure connection with the database system. All user passwords will be safeguarded using hashing algorithms to ensure data privacy and security.

Complete Security Lifecycle:

- Registration: Mandatory Email Verification.
- Data Storage: Passwords encrypted via Hashing.
- Authentication: Managed via Secure Sessions.
- Account Recovery: Secure Password Reset via email links.



A simple login/signup system by John and Uriel.



WORK PROGRESS

Back-end Development

HOME

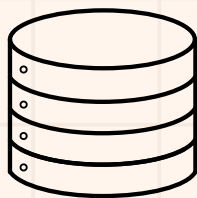
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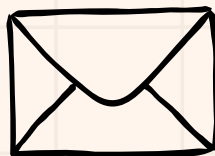
Speaker: Uriel Wuli

CORE INFRASTRUCTURE



Database (PHP + MySQL)

- Singleton connection
- User table as core



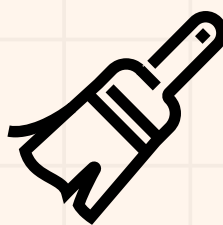
Mailer (PHPMailer)

- SMTP-based email
- Used for activation & password reset



Dependencies (JSON / Vendor)

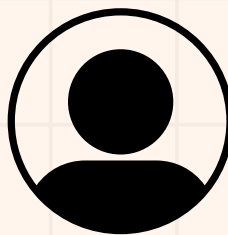
- Manage external libraries
- Support email & security



Global Styling (CSS)

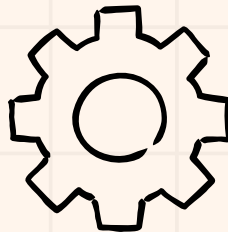
- Unified Brutalist UI
- Consistent design

USER REGISTRATION FLOW



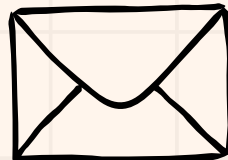
Signup Page (HTML + JS)

- Collect user data
- Real-time validation



Process Signup (PHP)

- Store user as inactive
- Generate secure token
- Send activation email



Signup Success (HTML)

- Prompt user to check email



Activate Account (PHP)

- Validate token
- Activate account

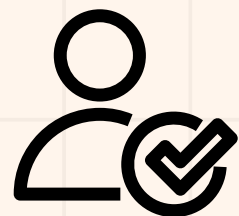


WORK PROGRESS

Back-end Development 

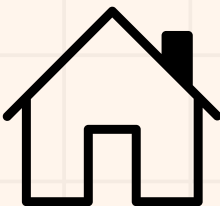
Speaker: Uriel Wuli

LOGIN & ACCESS CONTROL



Login Page

- Verify email & password
- Check activation



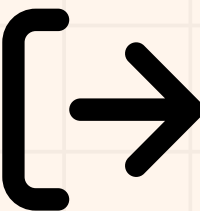
Index (Homepage)

- Only for authenticated users
- Block unauthorized access



Visitor Mode (Processing)

- Play without login
- No data saved



Logout (PHP)

- Destroy session
- Ensure security

PASSWORD RECOVERY FLOW



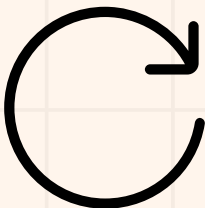
Forgot Password

- Enter email



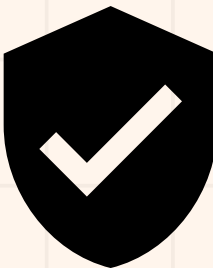
Send Reset (PHP)

- Generate time-limited token
- Send email



Reset Page (HTML)

- Input new password



Process Reset (PHP)

- Verify token
- Update password



TASK DESCRIPTION

Back-end Development

Search system

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Speaker: Uriel Wuli

PROPOSE

- Enable users to search player records and puzzle solutions
- Support querying historical results and gameplay data
- Improve user experience through fast data retrieval

CORE FUNCTIONS

- Connect to backend via RESTful API
- Query data from:
 - User records
 - Puzzle results
 - Gameplay history
- Return structured results to frontend for display



WORK PROGRESS

Back-end Development

Search system

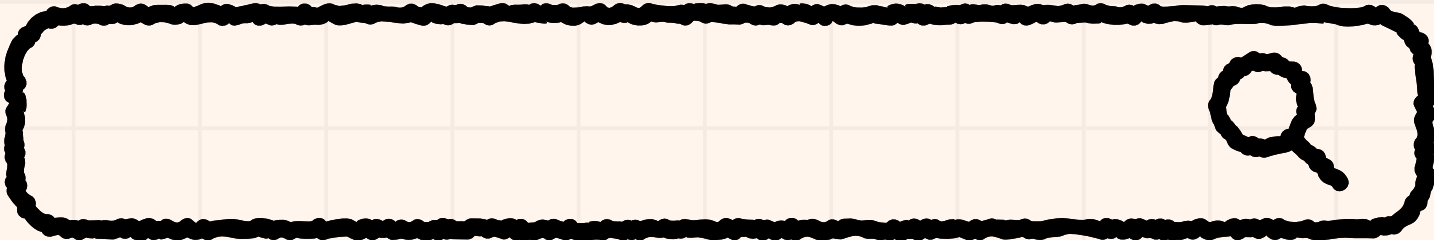


In Progress...

Waiting for database schema finalization

Depends on:

- User table structure
- Puzzle & result tables



TASK DISTRIBUTION

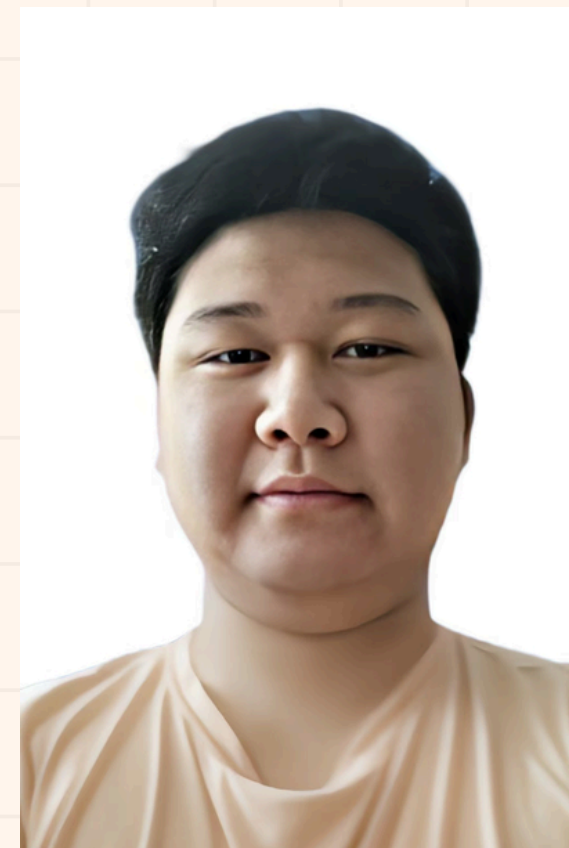
PAIR 11 - BACK-END DEVELOPMENT

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Speaker: Chen Joaquin Antonio



SEKI TOMOKI



CHEN JOAQUIN
ANTONIO

Responsible for managing the database system. (e.g. puzzle solutions, players and their records)

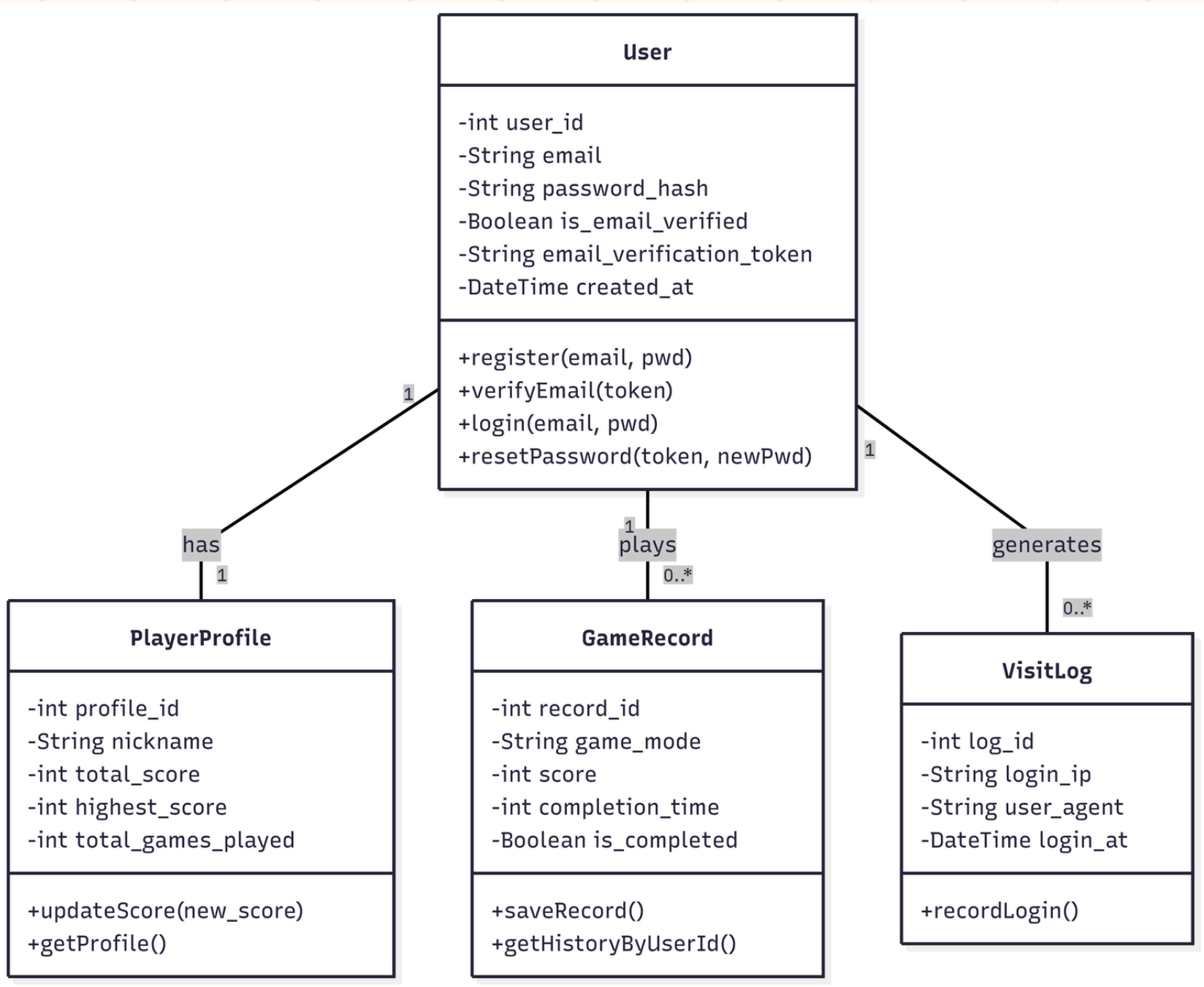
TASK DESCRIPTION

Database Management

Seki and Joaquin will be responsible for designing and implementing the core database system, including schemas, data operations, and audit logging.

Workflow Plan

- Define database schemas for core entities: User, PlayerProfile, GameRecord, and VisitLog, as shown in the class diagram.
- Implement game data operations: saving records, updating scores, retrieving profiles, and querying game history.
- Build the activity logging system to track user visits, including login IP, device info, and timestamps.
- Support user authentication and data queries, ensuring secure and efficient access to player and game data.

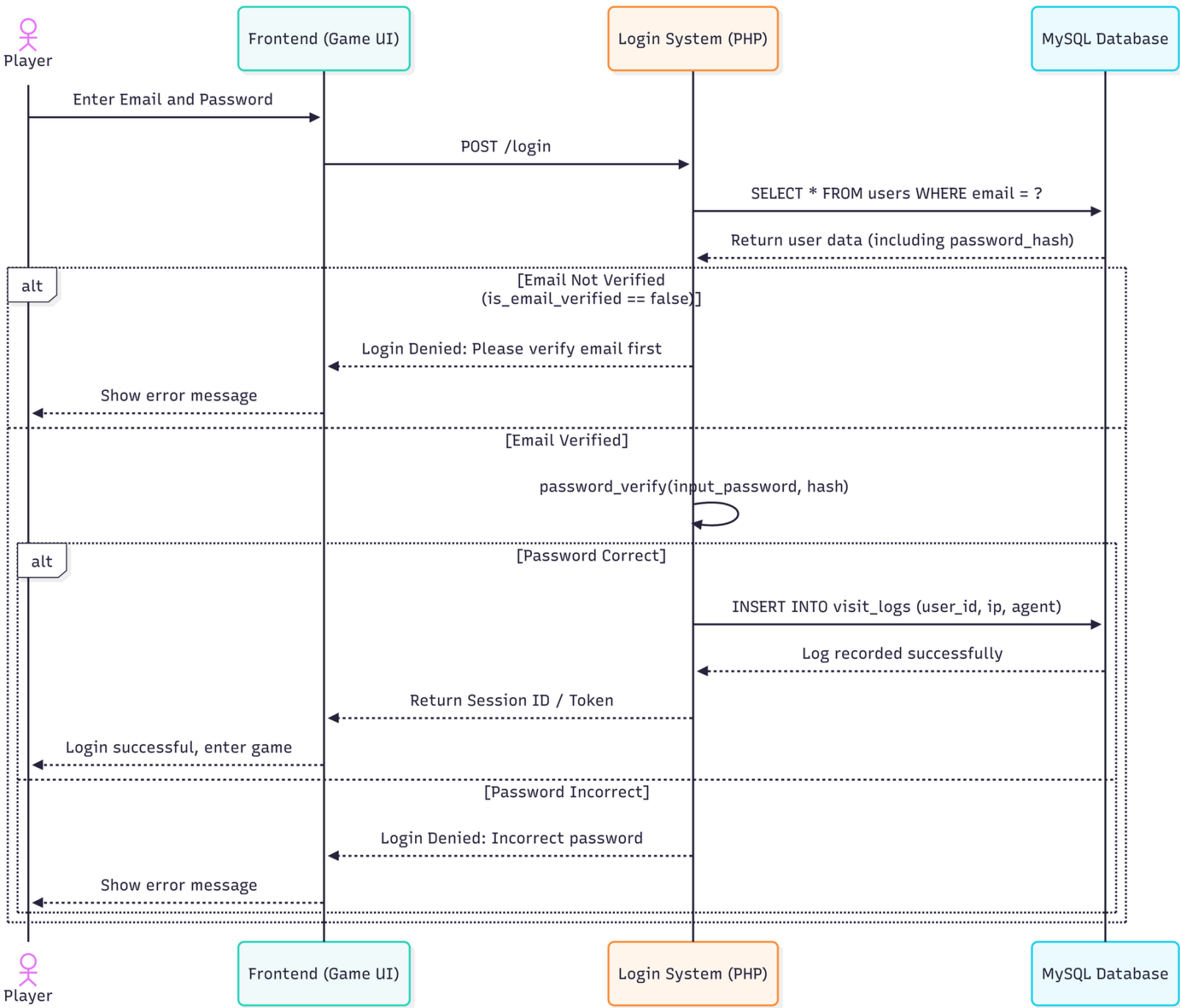


TASK DESCRIPTION

Database Management

This diagram details the secure login workflow for our platform:

- 1. **Request Submission:** The player enters credentials in the game UI, which sends a login request to the PHP backend.
- 2. **Database Lookup:** The backend retrieves the user’s account data, including password hash and verification status, from MySQL.
- 3. **Two-Step Validation:** The system first checks if the email is verified, then verifies the password using secure hashing.
- 4. **Activity Logging & Access:** Successful logins are recorded in visit_logs, and a session token is issued to grant the player access to the game.



WORK PROGRESS

Database Management

The core business database table structure design has been completed and successfully deployed to the cloud to support four core business scenarios:

USER

Supports the entire process of registration, login, email verification, and password reset to ensure account security.

GAME_RECORDS

Supports the entire process of registration, login, email verification, and password reset to ensure account security.

PLAYER_PROFILES

Links users and stores information such as nicknames, points, and game sessions, providing support for data statistics and presentation.

VISIT_LOGS

Records user login behaviors (compatible with IPv6), used for behavior tracing and security auditing.



WORK PROGRESS

Database Management

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Speaker: Chen Joaquin Antonio

Key Design Features

- **Foreign key association and cascading deletion:** Establish foreign key(such as user_id) relationships among all entities and maintain data consistency through the cascading deletion mechanism.
- **Password hash storage:** Hash the passwords before storing them to ensure the security of the credentials and match the verification logic in the login process.
- **Targeted index optimization:** Create indexes for frequently queried fields (such as scores, user history) to improve system performance.
- **Automated timestamp:** Utilize the automatic creation and update of timestamps to simplify the auditing tracking and data management processes.



CLOUD INFRASTRUCTURE

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Speaker: Seki Tomoki

ECS Console

CISC3003 Web Project Presentatio

阿里云-宝塔面板

47.239.186.224:8888 / localhost /

Login

https://ecs.console.aliyun.com/server/i-j6c987fqs74nlx08dgp/detail?regionId=cn-hongkong

阿里云

Workbench

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Main Account

Elastic Compute Service

Go to Standard Mode

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Security Groups

Elastic Network Interfaces

Block Storage

Snapshots

DNS

CloudMonitor

Expenses and Costs

All Menus

Instance

My Trial

Create Instance

Use your free trial credit to explore different configurations and find the right setup for your workload.

Validity Period: Apr 25, 2026, 19:00:00 - Jul 25, 2026, 19:00:00

Create Trial ECS

Release Trial ECS

Invite Others to Try

An analogy to help you understand the relationship between the validity period of the fre...

Step-by-step Guide to Understand the Free Trial

Remaining Free Trial Credit: 96.24%

Used Quota: 11.285 yuan

Free Quota: 300 yuan

Monthly Free Internet Traffic (Regions in mainland China): 100%

Used Quota: 0.000 GiB

Monthly Free Quota: 20 GiB

Estimated Hourly Credit Available: 79.91%

Used Quota: 0.167 yuan/hour

Free Quota: 0.833 yuan/hour

Monthly Free Internet Traffic (Regions outside mainland China): 99.98%

Used Quota: 0.032 GiB

Monthly Free Quota: 200 GiB

iZj6c987fqs74nlx08dgpZ

i-j6c987fqs74nlx08dgpZ

Running

China (Hong Kong) D

47.239.186.224 Public

Aliyun_64

2 Cores (vCPUs) 2 GiB

CPU Utilization 3.541%

Memory Usage Install Plug-in

Disk Usage Install Plug-in

Network Traffic Install Plug-in

Connect

Restart

Stop

Reset Password

Change Configurations

All Actions

Basic Information

Monitoring

Network and Security Group

Storage and Backup

Health Checks

Automated O&M

Security Protection

Instance Information

Health Status Normal

Billing Method Pay-as-you-go

Auto-release Time - Release

Instance Type ecs.e-c1m1.large

CPU and Memory 2 Cores (vCPUs) 2 GiB

Public IP Address 47.239.186.224

Primary Private IP Address 172.21.50.229

Internet Bandwidth 100 Mbps (Peak Bandwidth)

Billing Method for Network Usage Pay-by-traffic

Operating System Aliyun_64

Created Time Apr 25, 2026, 19:34:00

System Disk d-j6c5htod6tqr84m3254a

ECS Console

CISC3003 Web Project Presentatio

阿里云-宝塔面板

47.239.186.224:8888 / localhost /

Login

https://ecs.console.aliyun.com/image/region/cn-hongkong/imageInstance

阿里云

Workbench

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TOMOKISEKI

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Security Groups

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ECS / Images

Create Custom Image

Update Image

Copy Image

Import Image

Custom Images

Billing Note

Public Images

Shared Images

Marketplace Images

Community Images

Go to

宝塔Linux面板 (Alibaba Linux3操作系统/LAMP/LNMP/Java/Node/服务器管理/BT Panel)

Image ID

Status

Service Provider

Billing Method

Instance ID

Order ID

宝塔Linux面板 (Alibaba Linux3操作系统/LAMP/LNMP/Java/Node/服务器管理/BT Panel)

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In Use

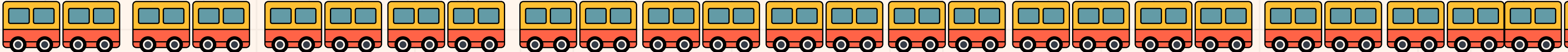
广东墨塔安全技术有限公司

Pay-As-You-Go

2026-04-25 22:28Create

i-j6c987fqs74nlx08dgp

285083577110685



SECURITY & ACCESSIBILITY

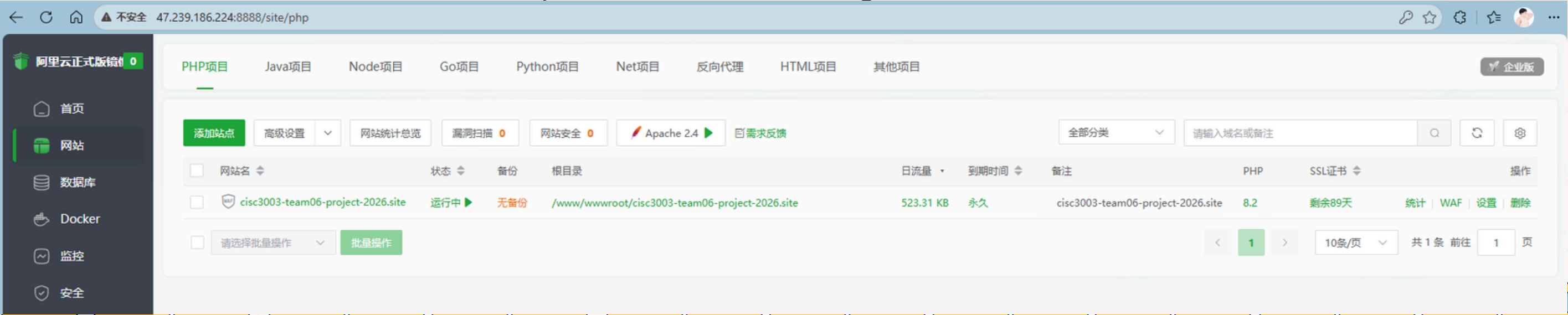
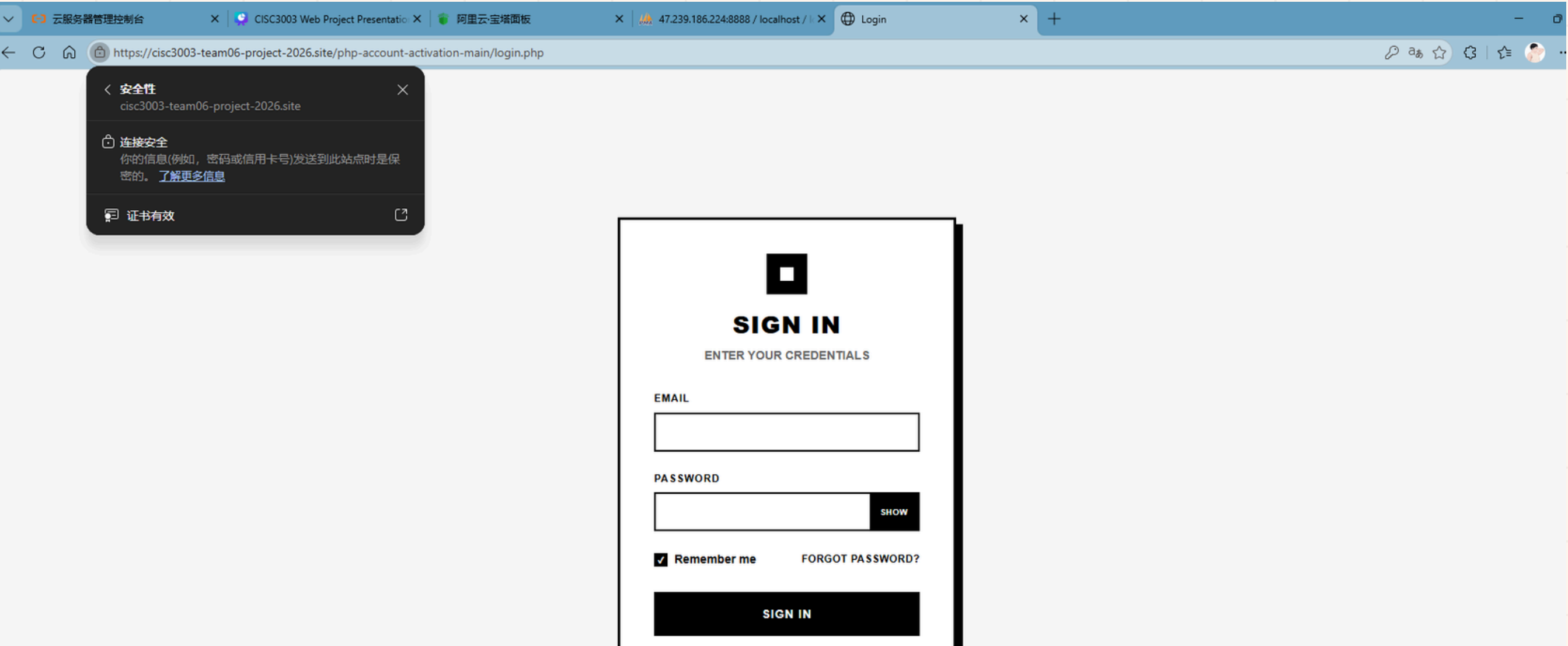
HOME

SERVICE

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Speaker: Seki Tomoki



THE DATA HUB

云服务器管理控制台

域名控制台

域名控制台

CISC3003 Web Project Presentatio

阿里云·宝塔面板

47.239.186.224:8888 / localhost /

Login

47.239.186.224:8888/phpmyadmin/index.php?route=/sql&pos=0&db=login_db&table=user

phpMyAdmin

近期访问 表收藏夹

新建

Information_schema

login_db

- 新建
- game_records
- player_profiles
- user
- visit_logs

服务器: localhost > 数据库: login_db > 表: user

浏览 结构 SQL 搜索 插入 导出 导入 操作 触发器

查询花费 0.0002 秒。)

解析 SQL [[创建 PHP 代码]] [刷新]

过滤行: 在表中搜索 按索引排序: 无

id	name	email	password_hash	account_activation_hash	reset_token_hash	reset_token_expires_at
5	URIEL	uriel341031@gmail.com	\$2y\$10\$JS4sEI8.pHqInGgshazBy.9PI8Tx7BF95KU2dJmlIH9...	NULL	237a0ce1fdb6c2c8ef457d8730881b7711b5931ce560b21900...	2026-04-28 16:08:39
6	antonio	dc22697@um.edu.mo	\$2y\$10\$ioIuh1NjyvyjfY6raNxUTObO4Iceqk3IjEffSVZWPn1...	661648c614339954e362d643a113efc8e0d294d10903309db3...	NULL	NULL
7	antonio	CHENJOAQUINANTONIO@163.com	\$2y\$10\$YEIN.XPVgMIBZhrYodp7bebm.BMeK1eb0V8QPv48IXD...	NULL	NULL	NULL
8	SEKI	sekitomoki20030219@gmail.com	\$2y\$10\$ZkKZyk9ZXKUKpVtCrNkzOO/9cFMb3J6dooPzC4FJD3N...	NULL	NULL	NULL



CHALLENGES

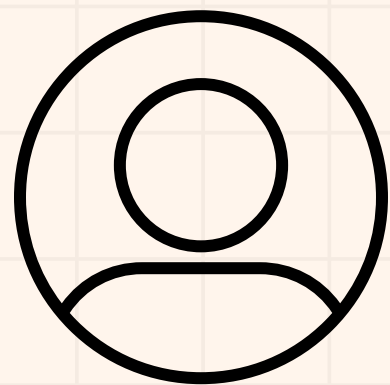
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Speaker: Collins, Chan Hio Tou

1. Integrating the **game logic** into the full-stack web framework smoothly.
2. Ensuring **secure user authentication**.
 - password hashing, email verification
 - avoiding common vulnerabilities (e.g. SQL injection and XSS attacks)
3. Achieving **cross-device responsiveness**.
4. Managing **real-time history tracking** without performance issues.
5. Designing **efficient search queries** for player records.

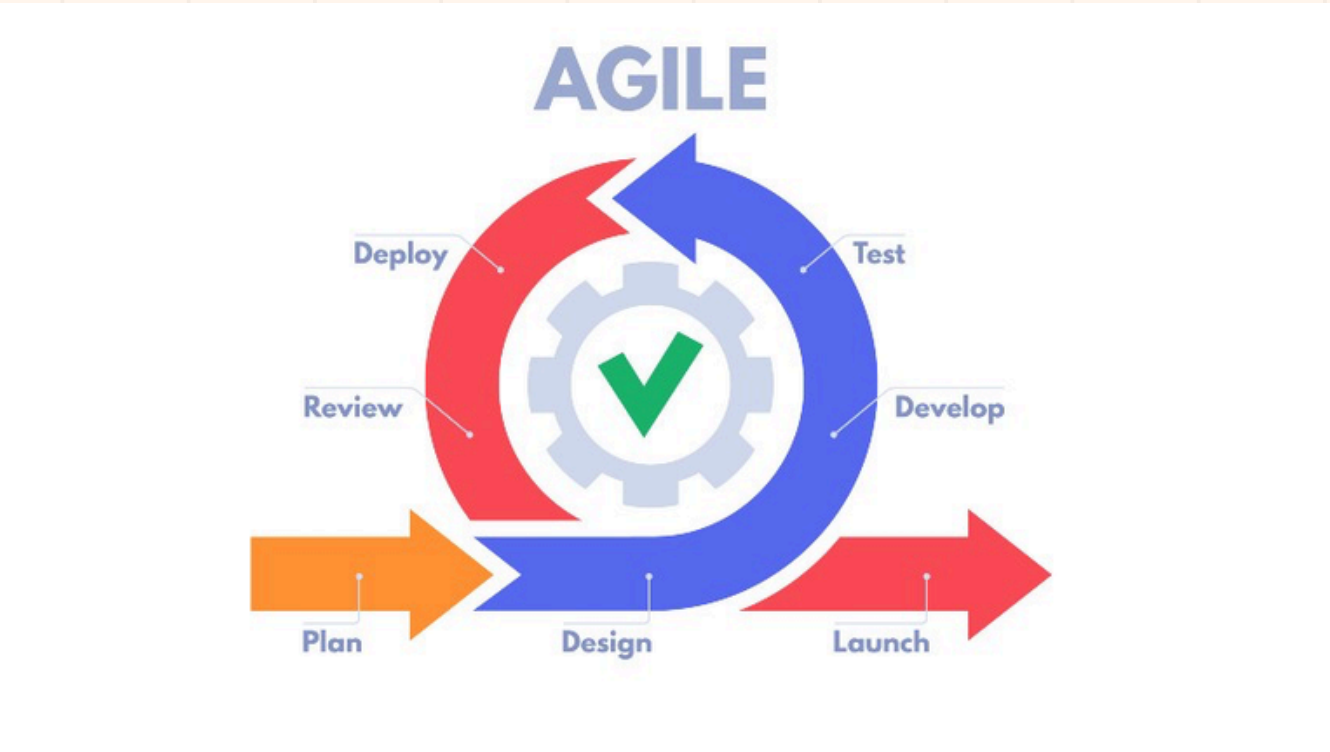


OTHER POSSIBLE CHALLENGES

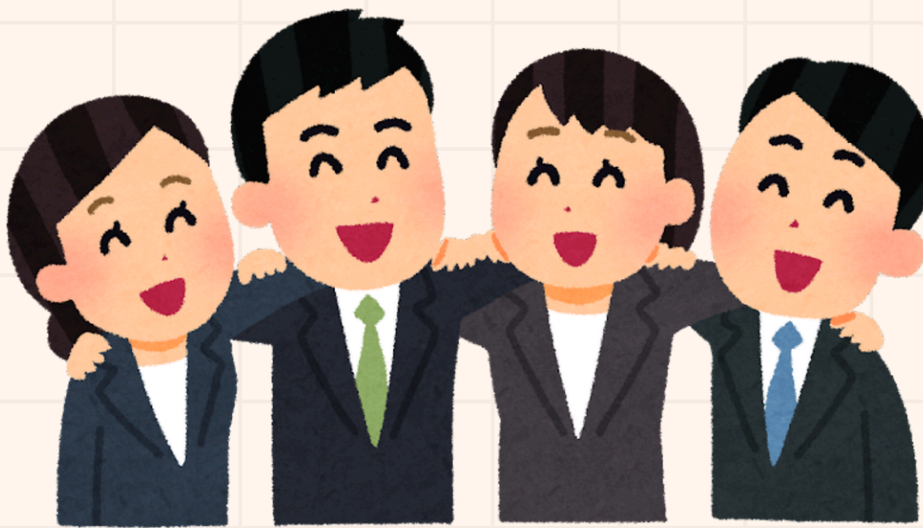


Team 03

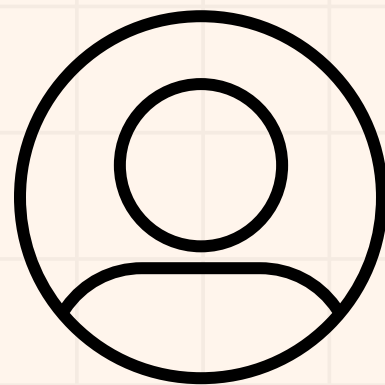
How will your team coordinate to integrate UI, APIs and database systems seamlessly?



An agile approach will be adopted. Continuous sprint meetings and integration testing will help in spotting mismatches between different system tiers.

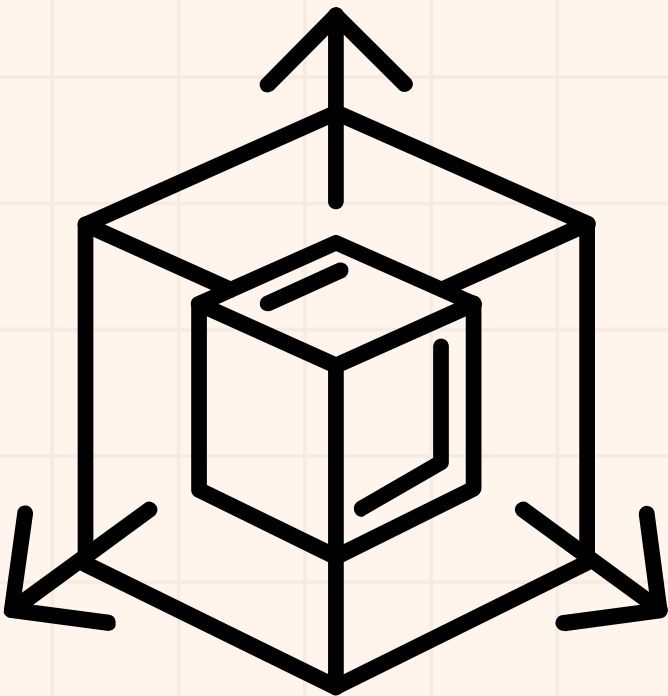


OTHER POSSIBLE CHALLENGES

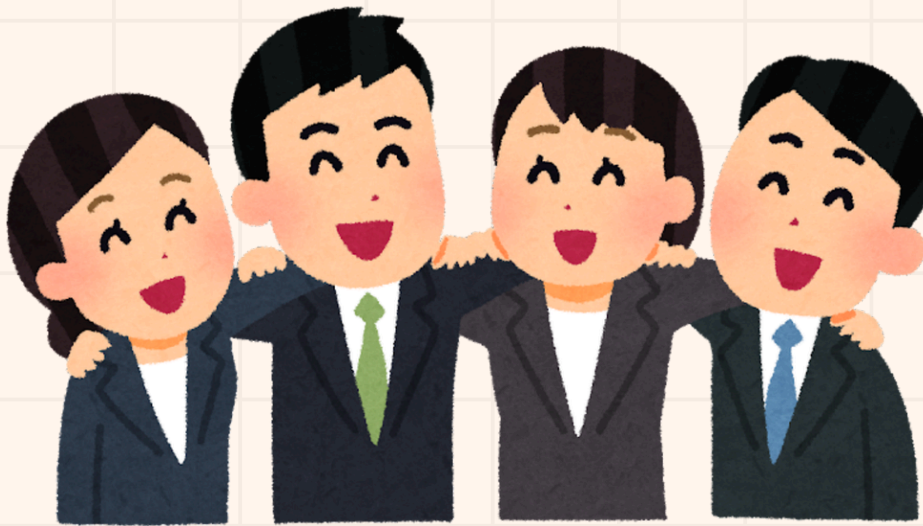


Team 02

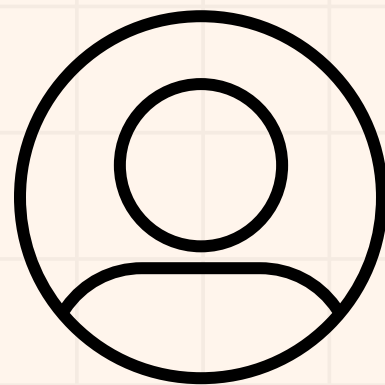
As your system scales, how will your team design such that the system can handle mass user load while maintaining smoothness of the game?



A layered architecture will be applied to distribute request pressure. The front-end ensures reactive design and asynchronous requests while the back-end can be scaled up to maintain responsiveness user high concurrency.

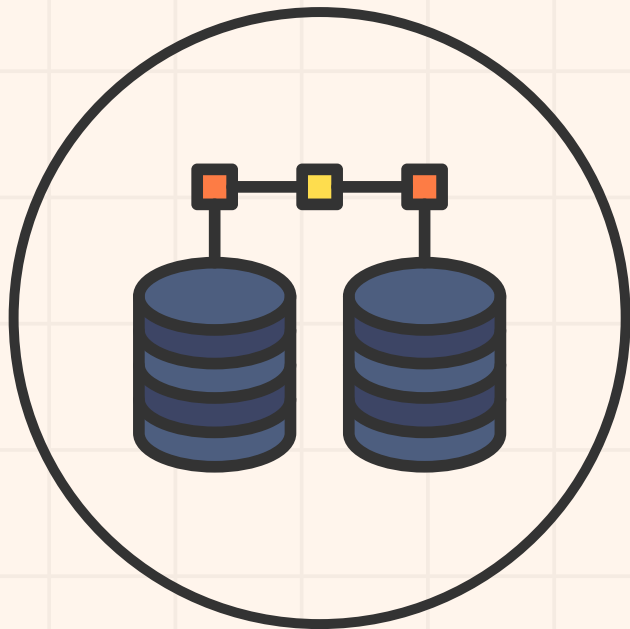


OTHER POSSIBLE CHALLENGES

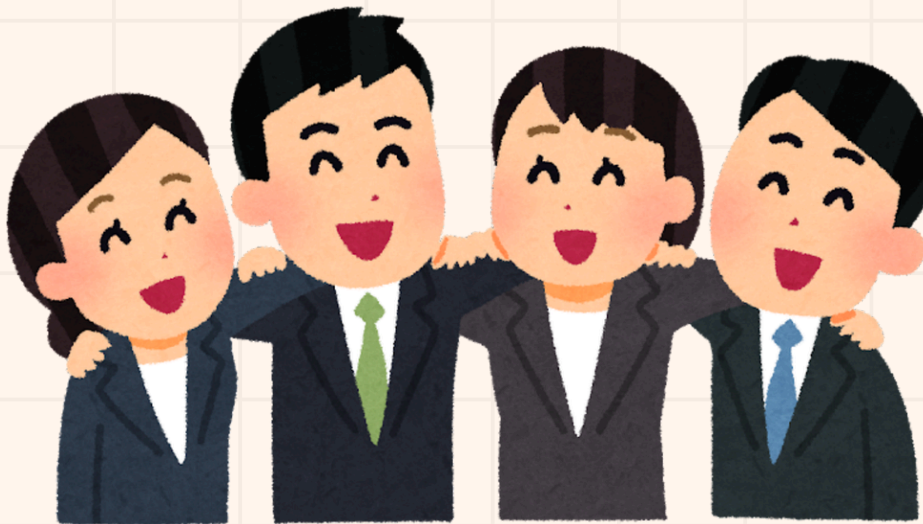


Team 05

How will you ensure fast search performance as the database grows?



We will use efficient indexing mechanisms and caching strategies. Also, when necessary, we will also separate player data from personal profiles.



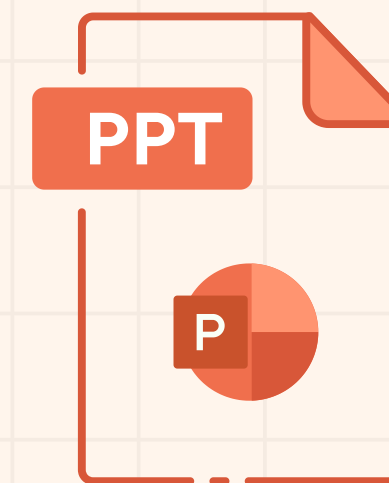
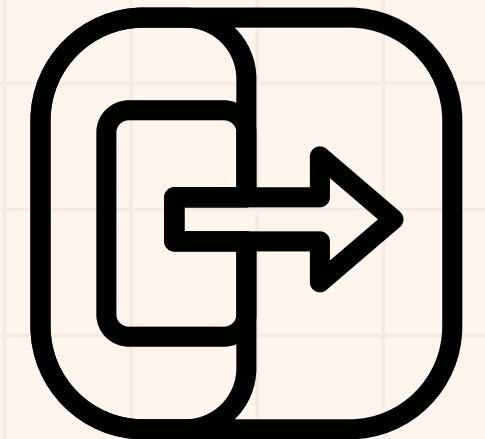
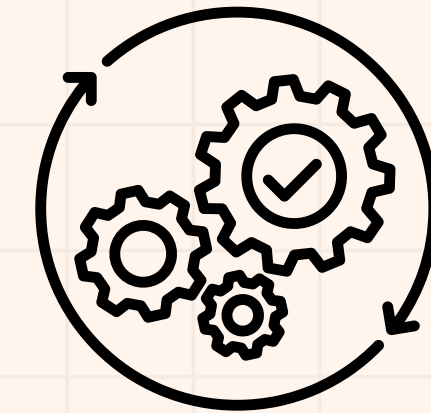
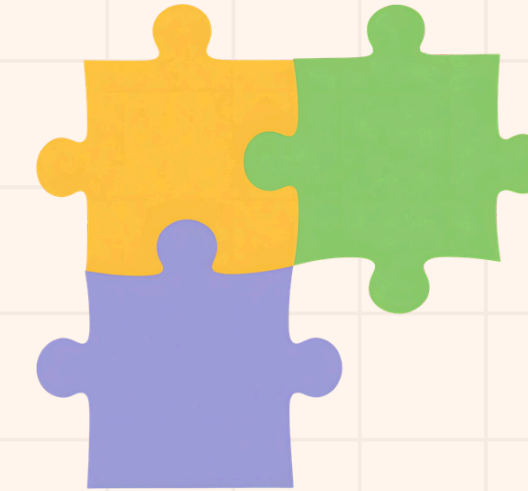
DELIVERABLES

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Speaker: Collins, Chan Hio Tou

Up to now, our deliverables for this project remain the same:

- Fully functional Train Puzzle Game
- Secure login/signup system with email verification
- Search service for records and solutions
- Visiting history and tracking
- Documentation (e.g. README, system design, user guide)
- Demo gameplay



CRITERIA OF SUCCESS



Responsive design, ensuring the game displays well on any device

Search function for player records and puzzle solutions, and clear visit logs.



Intuitive UI with smooth gameplay.

Secure authentication and password management.



Complete documentation and successful deployment.



EVALUATION

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Speaker: Collins, Chan Hio Tou

We are going to evaluate the application based on the following:

- **Content accuracy and relevance (35%)**
 - Does the project fully align with the proposal, with all features implemented correctly and relevant to course outcomes?
- **Technical implementation and functionality (35%)**
 - Does the application contain secure authentication, responsive design, search service, and history function, all running smoothly with no major bugs?
- **Design and user experience (20%)**
 - Is the interface intuitive, visually appealing? Is the game responsive across devices, with accessibility standards met? Is the game engaging and running smoothly?
- **Documentation completeness (10%)**
 - Does the project contain comprehensive documentation such as README, system architecture, E-R diagrams, user guide and demo visuals? Are they clear and professional?



PROJECT LEARNING

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Speaker: Collins, Chan Hio Tou



As an individual...

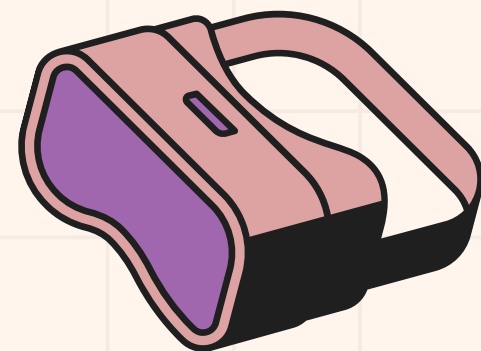
- Strengthen skills in full-stack development.
- Gain experience in implementing authentication, database design and responsive UI.
- Learn to integrate diverse components into one cohesive system.



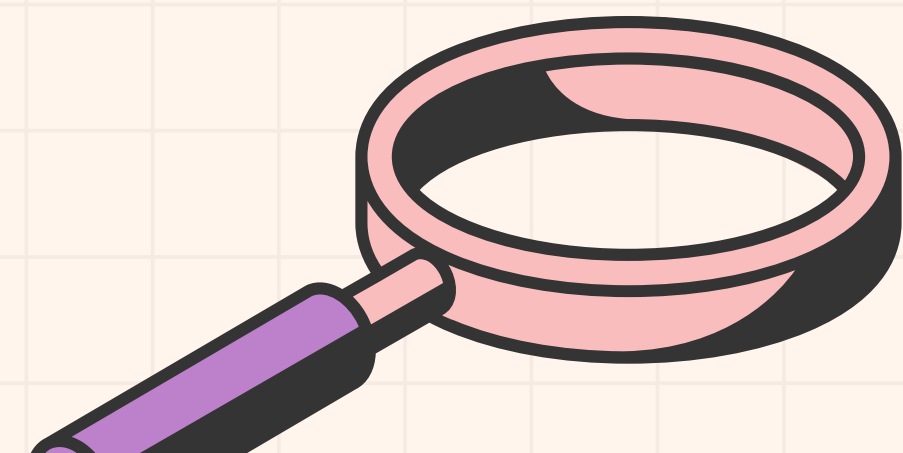
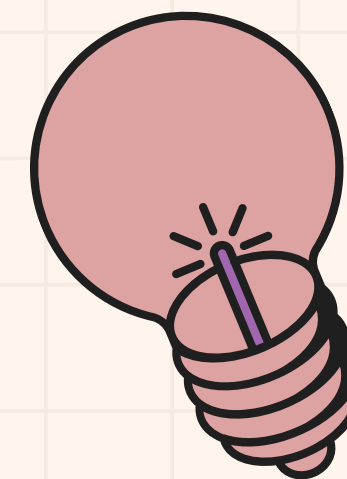
As a team...

- Practice agile collaboration and task distribution.
- Improve communication and project management skills.





~**DEMO TIME**~



REFERENCES

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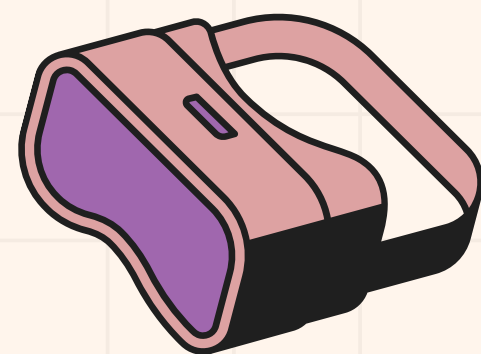
DECLARATION OF AI USAGE

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We declare that our web application is made with the help of the following AI models:

- Gemini 3.1 Pro – Used for code debugging, database SQL optimization, and generating mock data.
- Github Copilot – Provide assistance with HTML, CSS, JS, PHP and SQL code writing.





THANK YOU

